Intermediate-Complexity Coupled Toy Dynamics for DA Benchmarking

Case Study: Forcing-Corrupted Stochastic Lorenz-63 Setup

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1 Stochastic Lorenz-63 Testbed with Unobserved Forcing

To rigorously evaluate the failures of sequential data assimilation (DA) frameworks when exposed to both uncorrected background boundary drift and state-dependent subgrid volatility, we construct a low-dimensional benchmark system. This setup maps directly to the coupled dynamics challenges outlined in the state-space formulations of your proposal.

1.1 True Physical System: The Formulation

Inspired by the parameter and scale interaction coupling paradigms formalized in *Chapron et al. (2018)* (specifically following the structure of equations 30a–c), the true physical system is defined as a 4D continuous-time stochastic state-space. Here, the fluid components (X, Y, Z) undergo state-dependent (multiplicative) additive noise on the Y and Z rates of change, while being slowly driven by an unobserved, persistent land or boundary layer state (W_L):

$$\frac{\mathrm{d}X}{\mathrm{d}t} = \sigma(Y - X) + c_1 W_L \tag{1}$$

$$\frac{\mathrm{d}Y}{\mathrm{d}t} = X(\rho - Z) - Y + \sigma_0 Y \frac{\mathrm{d}B_{t,1}}{\mathrm{d}t} \tag{2}$$

$$\frac{\mathrm{d}Z}{\mathrm{d}t} = XY - \beta Z + \sigma_0 Z \frac{\mathrm{d}B_{t,2}}{\mathrm{d}t} \tag{3}$$

$$\frac{\mathrm{d}W_L}{\mathrm{d}t} = -\gamma(W_L - \bar{W}_L) + c_2 X + \sigma_L \frac{\mathrm{d}B_{t,3}}{\mathrm{d}t} \tag{4}$$

where:

- σ, ρ, β are the classical chaotic Lorenz parameters.
- W_L is the unobserved land or slow boundary driver with a relaxation timescale governed by $\gamma \ll 1$.
- c_1, c_2 dictate the bidirectional coupling strengths between the macroscale boundary layer and the fluid system.
- $\mathrm{d}B_{t,1}, \mathrm{d}B_{t,2}, \mathrm{d}B_{t,3}$ are mutually independent, standard one-dimensional Wiener increments.
- The terms $\sigma_0 Y \mathrm{d}B_{t,1}$ and $\sigma_0 Z \mathrm{d}B_{t,2}$ represent the *Chapron-inspired state-dependent additive noise*. Because the noise amplitude is directly proportional to the current magnitude of Y and Z, the stochastic volatility amplifies drastically as the trajectory flares outward into the lobes of the chaotic attractor, faithfully representing structural subgrid-scale atmospheric transitions.

1.2 Standard Numerical Calibration (Typical Parameter Values)

To implement this testbed as a valid twin experiment, the following parameter values are calibrated to maintain standard chaotic fluid regimes while ensuring visible scale interactions and forcing degradation:

Table 1: Calibrated Benchmark Parameter Configurations

Parameter Group	Symbol	Typical Value	Physical Interpretation
Fluid Attractor	σ	10.0	Prandtl number
	ρ	28.0	Classical Rayleigh number
	β	8/3	Attractor geometric factor
Time-Scale Separation	γ	0.05	Slow boundary relaxation rate ($\gamma \ll 1$)
	$\bar{W}_L$	0.0	Land mean equilibrium state
Cross-Component Fluxes	c_1	1.0	Fluid forcing sensitivity coefficient
	c_2	0.1	Atmospheric feedback footprint on land
Stochastic Variances	σ_0	0.08	Chapron state-dependent noise coefficient
	σ_L	0.20	Independent continuous boundary noise

1.3 The Operational System Deficit (The Twin DA Experiment)

To mimic real-world Earth System Modeling (ESM) limitations and test standard data assimilation algorithms, the operational predictive model (the code running inside the EnKF or variational window) is intentionally built with two critical structural flaws:

1. **Deterministic Physics Approximation:** The model completely omits the stochastic calculus terms, assuming a smooth, unperturbed deterministic representation of the fluid components ($\sigma_0 = 0$).

2. **Noisy Land Forcing Baseline:** The filter is blind to the online state evolution of the land W_L. Instead, the system is driven by a prescribed, corrupted external boundary dataset:
$$W_L^*(t) = W_L(t) + \eta(t) \tag{5}$$
where $\eta(t)$ is a colored, temporally auto-correlated noise process. This is typically modeled via an independent Ornstein-Uhlenbeck sequence with a decorrelation parameter $\tau_\eta = 5.0$ and variance $\sigma_\eta^2 = 0.5$.

1.4 Deconstructing the Sequential DA Failure Mode

When standard sequential filtering schemes (e.g., Ensemble Kalman Filters) are run on this setup, the joint impact of the uncorrected forcing drift $\eta(t)$ and the omitted state-dependent noise violates the core Markovian filtering assumptions:

- **The Collapse of the Stationary Model Error Assumption:** Traditional sequential filters require that model error (Q) be parameterized as a stationary, state-independent additive covariance matrix. In this configuration, however, the real-world error covariance is highly state-dependent and dynamic. It is minimized near the center of a lobe and explodes as the trajectory transitions into fast convective loops. Because the filter uses a static Q, its background error estimates are fundamentally misaligned with the true geometric state of the attractor.

- **Sequential Forcing Misattribution:** Suppose W_L^* carries a sustained bias η. This alters the path of X, while the fast fluctuations dynamically amplify the variance of Y and Z. When discrete fluid observations $y_{\text{OA}} = [X, Y, Z]^T$ are ingested at time t_k, the sequential filter computes an innovation. Operating step-by-step under the assumption of perfect boundary conditions (W_L^*), the Kalman Gain forces violent, unphysical adjustments onto the *fluid state variables* (X_k, Y_k, Z_k) to artificially minimize the current mismatch.

- **Attractor Disconnection:** By forcing state corrections onto the fluid to compensate for a hidden boundary bias and state-dependent subgrid noise, the filter frequently kicks the analyzed state entirely off the natural manifold of the chaotic attractor. This yields a classic "sawtooth" behavior: the analysis step pulls the model artificially toward observations, but the subsequent forecast step instantly experiences initialization shocks and diverges rapidly.

1.5 Direct Alignment with the Conditional Flow Matching Objective

This specific case study serves as an ideal verification framework for your proposed Conditional Flow Matching (CFM) objectives. Traditional DA methods collapse under this configuration because tracking a time-evolving, state-dependent non-Gaussian model error matrix $Q(\mathbf{x}_k)$ is mathematically intractable.

By contrast, the CFM approach trains a neural operator globally over the continuous timeline simultaneously:

$$\boldsymbol{\theta}^* \to \mathbb{E}(\mathbf{x}_{\text{OA}} \mid z_\tau, y_{\text{OA}}, W_L^*) \tag{6}$$

Because the vector field is optimized across the entire trajectory without collapsing the history into isolated, sequential assimilation blocks, the network learns to smooth over the high-frequency state-dependent volatility of the Y and Z components. It isolates the high-frequency subgrid noise from the slow, autocorrelated land forcing errors (η), mapping a clean, physically sound probability path that honors the true geometry of the stochastic attractor.

2 Case Study 2: Spatial Multi-Scale Lorenz-96 Testbed with Multiplicative Subgrid Noise

To investigate spatial multi-scale mechanics, cross-component spatial covariance propagation, and localized parameterization traps, we present an intermediate-complexity spatial lattice model. This setup directly maps to the coupled continuous-time formulations of your proposal, allowing an explicit benchmark of spatial misattribution and initialization shocks in sequential data assimilation (DA).

2.1 True Physical System: The Formulation

We structure a multi-component spatial lattice mapped along a single periodic latitudinal circle. The system governs a slow, large-scale **Ocean** variable (X_i). Nested within each discrete ocean sector are fast, chaotic **Atmospheric** variables ($Y_{j,i}$). Acting as a localized surface boundary with extended temporal memory is the **Land/Soil Moisture** layer ($Z_{k,i}$).

To remain mathematically consistent with the scale-interaction principles of *Chapron et al. (2018)*, we parameterize the unresolved fast atmospheric transitions or convective eddies as a localized, state-dependent (multiplicative) additive noise term embedded directly inside the atmospheric rate equations:

$$\textbf{Ocean } (x_O): \quad \frac{\mathrm{d}X_i}{\mathrm{d}t} = -X_{i-1}(X_{i-2} - X_{i+1}) - X_i + F_O - \frac{h_A c_A}{b_A} \sum_{j=1}^{J} Y_{j,i} + c_L Z_{k,i} \quad (7)$$

$$\textbf{Atmosphere } (x_A): \quad \frac{\mathrm{d}Y_{j,i}}{\mathrm{d}t} = -b_A Y_{j+1,i}(Y_{j+2,i} - Y_{j-1,i}) - Y_{j,i} + F_A + \frac{h_A c_A}{b_A} X_i + \alpha_L Z_{k,i} + \sigma_0 Y_{j,i} \frac{\mathrm{d}B_{j,i}^{(1)}}{\mathrm{d}t} \quad (8)$$

$$\textbf{Land } (x_L): \quad \frac{\mathrm{d}Z_{k,i}}{\mathrm{d}t} = -\gamma Z_{k,i} + d_A \left(\sum_{j=1}^{J} Y_{j,i} \right) + \sigma_L \frac{\mathrm{d}B_{k,i}^{(2)}}{\mathrm{d}t} \quad (9)$$

where:

- $i = 1, \ldots, N_O$ represents the large-scale ocean sectors positioned across the spatial grid.
- $j = 1, \ldots, J$ maps the highly chaotic atmospheric grid nodes nested hierarchically within each specific ocean sector i.
- $Z_{k,i}$ structures the localized land boundary grid patches matching the lattice blocks, operating with a slow relaxation timescale ($\gamma \ll 1$).
- $\mathrm{d}B_{j,i}^{(1)}$ and $\mathrm{d}B_{k,i}^{(2)}$ are mutually independent, standard one-dimensional spatial Wiener increments.
- The stochastic expression $\sigma_0 Y_{j,i} \mathrm{d}B_{j,i}^{(1)}$ enforces a localized, state-dependent additive noise. When an atmospheric wave peaks locally ($Y_{j,i}$ is large), the unrepresented subgrid-scale turbulent fluctuations amplify proportionally, capturing real-world energy dissipation profiles.

2.2 Standard Numerical Calibration (Typical Parameter Values)

To ensure clear scale separation while keeping all components deeply embedded within their fully chaotic, non-linear regimes, we adopt the following standard parameter settings for the twin experiments:

Table 2: Calibrated Spatial L96 Configuration Parameters

Parameter Group	**Symbol**	**Typical Value**	**Physical / Spatial Interpretation**
Lattice Dimensions	N_O	8 to 36	Large-scale macro oceanic sectors
	J	4 to 10	Atmospheric nodes nested per ocean sector
Core Fluid Forcing	F_O, F_A	8.0	Uncoupled baseline chaotic forcing inputs
	b_A	10.0	Atmospheric advective scale-acceleration factor
	h_A, c_A	1.0	Ocean-atmosphere linear coupling constants
Land Boundary Layer	γ	0.03	Slow soil moisture memory relaxation rate
	c_L	0.5	Thermal/drainage ocean impact coefficient
	α_L	0.8	Land boundary layer forcing on atmosphere
	d_A	0.1	Integrated precipitation feedback onto land

3 Case 3: Spatial Multi-Layer Shallow Water System with Multiplicative Hydrodynamic Noise

To challenge data assimilation architectures with realistic hydrodynamic invariants, conservation properties, and explicit non-linear wave interactions (e.g., gravity waves, Rossby waves), we introduce a 2D Partial Differential Equation (PDE) benchmark system. This system is structured directly around the grand-state interaction principles and directional flux conventions defined in Section 1.1 of your proposal notes.

3.1 True Physical System: Component-Wise Fluid Formulation

The true continuous-time state is distributed across a 2D horizontal periodic domain. In full alignment with Equation (1) of the proposal, the system is partitioned into an Atmosphere fluid component (x_A), an Ocean fluid component (x_O), and a Land boundary state (x_L).

We govern the fluid layers by horizontal velocity vector fields $\mathbf{u}_c = (u_c, v_c)$ and total layer mass thicknesses h_c. To integrate the scale-interaction paradigms of *Chapron et al. (2018)*, we insert a state-dependent, multiplicative stochastic noise matrix (Σ_A) directly into the atmospheric momentum channels to represent unrepresented subgrid wind gust volatility:

$$\textbf{Atmosphere } (x_A): \quad \frac{\partial \mathbf{u}_A}{\partial t} = \mathcal{M}_{\mathbf{u}_A}(\mathbf{u}_A, h_A) + \mathcal{F}_{O\to A}(\mathbf{u}_O, \mathbf{u}_A) + \mathcal{F}_{L\to A}(q_L, \mathbf{u}_A) + \boldsymbol{\sigma}_0 \|\mathbf{u}_A\| \dot{\mathbf{W}}_A \tag{10}$$

$$\frac{\partial h_A}{\partial t} = \mathcal{M}_{h_A}(\mathbf{u}_A, h_A) + \mathcal{F}_{L\to h_A}(q_L, \mathbf{u}_A) - \mathcal{F}_{h_A\to L}(h_A) \tag{11}$$

$$\textbf{Ocean } (x_O): \quad \frac{\partial \mathbf{u}_O}{\partial t} = \mathcal{M}_{\mathbf{u}_O}(\mathbf{u}_O, h_O) + \mathcal{F}_{A\to O}(\mathbf{u}_A, \mathbf{u}_O) \tag{12}$$

$$\frac{\partial h_O}{\partial t} = \mathcal{M}_{h_O}(\mathbf{u}_O, h_O) + \mathcal{F}_{L\to h_O}(q_L) \tag{13}$$

$$\textbf{Land } (x_L): \quad \frac{\partial q_L}{\partial t} = \mathcal{F}_{h_A\to L}(h_A) - \mathcal{F}_{L\to h_A}(q_L, \mathbf{u}_A) - \mathcal{F}_{L\to h_O}(q_L) + \sigma_L \dot{W}_L \tag{14}$$

3.2 Deconstruction of Internal Operators ($\mathcal{M}_c$) and Coupling Fluxes ($\mathcal{F}_{c\to c'}$)

The internal forward physics models ($\mathcal{M}_c$) representing classic shallow water hydrodynamics are structured as:

$$\mathcal{M}_{\mathbf{u}_c}(\mathbf{u}_c, h_c) = -(\mathbf{u}_c \cdot \nabla)\mathbf{u}_c - f\mathbf{k} \times \mathbf{u}_c - g\nabla\mathcal{H}_{\text{total}} \tag{15}$$

$$\mathcal{M}_{h_c}(\mathbf{u}_c, h_c) = -\nabla \cdot (h_c \mathbf{u}_c) \tag{16}$$

where f is the Coriolis parameter, g is gravity, $\mathbf{k}$ is the vertical unit vector, and the total hydrostatic pressure head gradient evaluates to $\mathcal{H}_{\text{total}} = h_A + h_O$ for the atmosphere, and $\mathcal{H}_{\text{total}} = h_O$ for the ocean layer.

The directional cross-component coupling flux operators ($\mathcal{F}_{c\to c'}$) are explicitly defined as:

- **Atmosphere-to-Ocean Wind Stress ($\mathcal{F}_{A\to O}$):** Quadratic bulk aerodynamic drag equation transferring momentum to the ocean surface:

$$\mathcal{F}_{A\to O}(\mathbf{u}_A, \mathbf{u}_O) = C_D \|\mathbf{u}_A - \mathbf{u}_O\| (\mathbf{u}_A - \mathbf{u}_O) \tag{17}$$

- **Ocean-to-Atmosphere Surface Friction ($\mathcal{F}_{O\to A}$):** The equal and opposite mechanical drag experienced by the lower atmospheric boundary layer:

$$\mathcal{F}_{O\to A}(\mathbf{u}_O, \mathbf{u}_A) = -C_D\|\mathbf{u}_A - \mathbf{u}_O\|(\mathbf{u}_A - \mathbf{u}_O) \tag{18}$$

- **Land-to-Atmosphere Latent Heat Evaporation ($\mathcal{F}_{L\to h_A}$):** State-dependent moisture mass transport scaling linearly with wind speed and available soil wetness:

$$\mathcal{F}_{L\to h_A}(q_L, \mathbf{u}_A) = C_E q_L\|\mathbf{u}_A\| \tag{19}$$

- **Atmosphere-to-Land Precipitation ($\mathcal{F}_{h_A\to L}$):** A threshold-based rain generator that deposits mass when local atmospheric water thickness limits are breached:

$$\mathcal{F}_{h_A\to L}(h_A) = R_0 \max(0, h_A - h_{\text{crit}}) \tag{20}$$

- **Land-to-Ocean Basin Runoff ($\mathcal{F}_{L\to h_O}$):** Localized overland hydraulic drainage feeding groundwater excesses straight into coastal ocean boundaries:

$$\mathcal{F}_{L\to h_O}(q_L) = C_R \max(0, q_L - q_{\text{sat}}) \tag{21}$$

- **Land-to-Atmosphere Momentum Roughness ($\mathcal{F}_{L\to A}$):** Boundary layer mechanical resistance dampening wind vectors over land masses:

$$\mathcal{F}_{L\to A}(q_L, \mathbf{u}_A) = -C_L q_L \mathbf{u}_A \tag{22}$$

3.3 Standard Numerical Calibration (Typical Parameter Values)

To ensure physical stability across the 2D spatial grid while preserving characteristic scale separations, the twin experiment utilizes the following calibrated physical baseline configurations:

Table 3: Calibrated Spatial Multi-Layer PDE Configurations

Parameter Group	Symbol	Typical Value	Physical Interpretation / Domain Scaling
Domain & Mesh	Ω	64×64	2D periodic grid resolution spatial layout
Fluid Physics	g	9.81 m/s^2	Normalized gravitational acceleration constant
	f	10^{-4} s^{-1}	Mid-latitude f-plane Coriolis parameter
Mean Heights	$\bar{h}_A$	5000 m	Baseline atmospheric fluid layer mass thickness
	$\bar{h}_O$	200 m	Baseline ocean fluid layer mass thickness
Boundary Coupling	C_D	1.5×10^{-3}	Surface momentum drag transfer coefficient
	C_E	1.0×10^{-3}	Ground latent heat evaporation coefficient
	C_R	1.0×10^{-2}	Overland basin hydraulic runoff rate
	q_{sat}	1.0	Soil moisture volumetric saturation limit
Noise Elements	$\boldsymbol{\sigma}_0$	0.04	Chapron atmospheric wind stress scaling coefficient
	σ_L	0.05	Slow integrated background soil moisture noise

3.4 Deconstructing the Adjoint Shock in Variational DA

When standard variational algorithms (such as strong- or weak-constraint 4DVar) attempt to compute fluid trajectories over this system while omitting the land storage layer ($q_L = 0$), they encounter a fatal state-dependent model error trap:

1. **The State-Dependent Covariance (Q) Disconnection:** In nature, heavy rain saturates the land, which slowly feeds water into the ocean and moisture into the air for days. Because the operational model lacks this land variable ($\theta_U = 0$), its simulated evaporation and runoff instantly drop to zero when the storm clears.

2. **Variational Misattribution:** When fluid observations y_{OA} display sustained localized fluid deviations, the 4DVar gradient descent loop $\nabla_{\mathbf{x}_{\mathrm{OA}}}\mathcal{J}$ attempts to resolve the error. Lacking any representation of historical boundary layer memory, it is mathematically forced to misattribute the continuous mass and vapor entries entirely to fluid initial states.

3. **The Adjoint Shock Execution:** To force the land-free, deterministic shallow-water equations to match observed high ocean levels and moisture paths, the adjoint model forces *highly turbulent, non-physical velocity gradients* ($\nabla \cdot \mathbf{u}_A, \nabla \cdot \mathbf{u}_O$) into the initial fluid structures. The moment the forecast begins, these artificial divergence zones trigger massive, unphysical gravity wave shocks that radiate across the grid, destroying mass conservation properties and causing forecast skill to collapse exponentially.

3.5 Resolution via Conditional Flow Matching (CFM)

This multi-layer shallow water system provides the ultimate justification for **Section 4.2** of your proposal notes. Tracking or specifying a state-dependent, non-Gaussian model error covariance path matrix $Q(\mathbf{x}, \mathbf{x}', t)$ across a 2D grid in a traditional sequential DA setup is computationally impossible.

A **Conditional Flow Matching (CFM)** operator resolves this issue. By optimizing the continuous optimal transport velocity field over the entire space-time sequence volume simultaneously ($\mathbb{E}(x_{\mathrm{OA}} \mid z_\tau, y_{\mathrm{OA}}, x_L^*, \theta^*)$), the neural operator does not rely on step-by-step Markovian propagation. It implicitly learns to recognize the space-time footprints left behind by unobserved boundaries, mapping clean probability paths that honor the underlying fluid manifolds and shallow-water invariants without ever generating artificial velocity shocks.

4 Algorithmic Setup: Scoring Rules and Operational Operator Distances

To establish a principled methodology to decipher how modeling assumptions considered in DA schemes drive the representation of posteriors, we map both the generative CFM model and the sequential EnKF into a common continuous probability path space. This permits the calculation of explicit proper scoring rules as distances between flow matching operators.

4.1 Continuous Vector-Field Mapping for the EnKF

To compare the sequential EnKF updates against a continuous CFM operator, we must re-interpret the discrete EnKF state corrections as an implicit velocity field mapping across an artificial inner interpolation timeline $\tau \in [0, 1]$ at analysis time t_k.

Let the EnKF background (prior) ensemble be $\mathbf{E}_k^b = \{\mathbf{x}_{n,k}^b\}_{n=1}^N$ and the analysis (posterior) ensemble be $\mathbf{E}_k^a = \{\mathbf{x}_{n,k}^a\}_{n=1}^N$. We define a linear deterministic optimal transport path for each individual ensemble member:

$$\mathbf{x}_{n,k}(\tau) = (1 - \tau)\mathbf{x}_{n,k}^b + \tau \mathbf{x}_{n,k}^a \tag{23}$$

The empirical velocity field generated by this instantaneous sequential EnKF correction is the constant displacement vector:

$$\mathbf{v}_{\mathrm{EnKF}}(\mathbf{x}_{n,k}(\tau), \tau) = \mathbf{x}_{n,k}^a - \mathbf{x}_{n,k}^b \tag{24}$$

By contrast, the trained CFM operator $\mathbf{u}_{\boldsymbol{\theta}}(\mathbf{x}, \tau \mid y_{\mathrm{OA}}, W_L^*)$ directly outputs a smooth, time-continuous vector field that pushes a simple base distribution toward the true, non-Gaussian posterior trajectory.

4.2 Metric A: Probability Path Divergence ($\mathcal{D}_{\mathbf{flow}}$)

Evaluating over a verification trajectory window of length K, we can compute the direct functional L_2 distance between the EnKF displacement paths and the CFM probability path maps:

$$\mathcal{D}_{\text{flow}}(\text{CFM}, \text{EnKF}) = \frac{1}{K \cdot N} \sum_{k=1}^{K} \sum_{n=1}^{N} \int_0^1 \left\| \mathbf{u}_{\boldsymbol{\theta}}(\mathbf{x}_{n,k}(\tau), \tau \mid y_{\mathrm{OA}}, W_L^*) - \left(\mathbf{x}_{n,k}^a - \mathbf{x}_{n,k}^b\right) \right\|_2^2 \mathrm{d}\tau \quad (25)$$

When the unobserved boundary bias $\eta(t)$ or state-dependent subgrid volatility forces the true posterior path to curl away into highly non-Gaussian, multi-modal structures, $\mathcal{D}_{\text{flow}}$ will sharply spike. This provides a direct, quantitative metric showing where the linear assumptions of sequential DA collapse compared to the true stochastic geometry.

4.3 Metric B: Multivariate Energy Score Distance ($\mathcal{D}_{\mathbf{energy}}$)

To measure posterior discrepancy at the final integrated state space ($\tau = 1$) without depending on the inner interpolation path, we deploy the multivariate Energy Score—a strict proper scoring rule. Let $\mathbf{X}_k, \mathbf{X}_k' \sim p_{\text{CFM}}$ be independent posterior samples obtained by integrating the CFM vector field via an ODE solver, and let $\mathbf{Y}_k, \mathbf{Y}_k' \sim p_{\text{EnKF}}$ be independent samples drawn directly from the operational EnKF analysis ensemble:

$$\mathcal{D}_{\text{energy}}(\text{CFM}, \text{EnKF}) = \frac{1}{K} \sum_{k=1}^{K} \left[2\,\mathbb{E}\left[\|\mathbf{X}_k - \mathbf{Y}_k\|_2\right] - \mathbb{E}\left[\|\mathbf{X}_k - \mathbf{X}_k'\|_2\right] - \mathbb{E}\left[\|\mathbf{Y}_k - \mathbf{Y}_k'\|_2\right]\right] \quad (26)$$

Because the Energy Score is maximized when probability distributions diverge, any spatial initialization shock, mass imbalance, or manifold disconnection suffered by the sequential EnKF will drive $\mathcal{D}_{\text{energy}}$ upwards, providing a clean baseline to chart performance as a function of land memory (γ) and stochastic flaring (σ_0).

References

1. Chapron et al (2018). SLarge-scale flows under location uncertainty: a consistent stochastic framework. *QJRMS*.